

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026
Comprehensive Examination

EC-3 Regular

Course No. : AIMLCZC416
Course Title : Mathematical Foundations for Machine Learning
Nature of Exam : Open Book
Weightage : 40%
Duration : 2 hrs 30 mts
Date of Exam : 01-03-2026

No. of Pages = 3

No of questions = 5

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.
4. Notation used here are as per the text book.

Q1. Consider the matrix $A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$.

(a) Find all the eigen values and eigen vectors of A. [5M]

(b) Using (a) find a diagonal matrix D and an invertible matrix P such that $AP = PD$. [2M]

(c) Find the singular values of the matrix $\begin{bmatrix} \sqrt{3} & \frac{1}{\sqrt{3}} & 0 \\ 0 & \sqrt{\frac{8}{3}} & 0 \\ 0 & 0 & 2 \end{bmatrix}$. [1M]

Q2. Let $v_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ -1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ 2 \\ 3 \\ 4 \end{bmatrix}$, $v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, and $v_4 = \begin{bmatrix} -6 \\ 3 \\ -3 \\ -3 \end{bmatrix}$,

(a) Determine whether the vectors v_1, v_2, v_3, v_4 are linearly dependent or linearly independent in R^4 . Justify your answer. [3M]

(a) Prove that $U = \text{span} \{v_1, v_2, v_3, v_4\}$ is a subspace of R^4 . Also find a basis and dimension of U. [3M]

(c) What is the rank of the matrix $B = \begin{bmatrix} -3 & 0 & 0 & -6 \\ 1 & 2 & 1 & 3 \\ 0 & 3 & 0 & -3 \\ -1 & 4 & 0 & -3 \end{bmatrix}$. Justify your answer. Also find the determinant of the matrix B. [2M]

Q3. Let $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.

An inner product on $\mathbb{R}^2$ is defined by $\langle u, v \rangle = u^T A v$, for $u, v \in \mathbb{R}^2$.

The norm induced by this inner product is $\|x\| = \sqrt{\langle x, x \rangle}$.

Assume that a signal tower is located at the origin $(0, 0)$. The cost of placing a receiver at a point $(x, y) \in \mathbb{R}^2$ is given by $C(x, y) = 2 \| (x, y) \|^2$. Suppose the receiver must be placed in the region

$$R = \{(x, y) \in \mathbb{R}^2 : x + y - 1 \geq 0, x + y - 3 \leq 0\}.$$

(a) State and prove the necessary properties of the matrix A such that $\langle u, v \rangle = u^T A v$ defines an inner product on $\mathbb{R}^2$. [2M]

(b) Formulate the constrained optimization problem to determine the point in the region R at which the cost function $C(x, y)$ is minimized. [1M]

(c) Let $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ be the Lagrange multipliers corresponding to the constraint $x + y - 1 \geq 0$ and $x + y - 3 \leq 0$ respectively. Formulate the Lagrangian function and hence write the corresponding Lagrangian dual problem. [2M]

(d) Verify that the Karush–Kuhn–Tucker (KKT) conditions are satisfied at $(x, y, \lambda_1, \lambda_2) = \left(\frac{1}{2}, \frac{1}{2}, 8, 0\right)$. [3M]

Q4. A retail shop wants to predict their sales corresponding to the money they spent on advertisement using the following data. A linear regression model with bias is assumed:

$$\hat{y}_i = w_0 + w_1 x_i$$

Money spent for advertisement in ten thousands (x)	Sales in lacs (y)
5	3
6	5

The model parameters are learned by minimizing the Squared Error loss function $J(w_0, w_1) = \frac{1}{2} \sum_{i=1}^2 (\hat{y}_i - y_i)^2$ using gradient descent.

(a) Formulate the Squared Error loss function $J(w_0, w_1)$ explicitly for the given dataset. Compute the Hessian matrix of J with respect to w_0 and w_1 , and use it to prove that J is a convex function of the parameters (w_0, w_1) . [3M]

(b) Compute the gradient of the Squared Error loss function: $\nabla J = \begin{bmatrix} \frac{\partial J}{\partial w_0} \\ \frac{\partial J}{\partial w_1} \end{bmatrix}$. [1M]

(c) Taking the initial parameter values $w_0^{(0)} = 0$, $w_1^{(0)} = 0$, perform two iterations of gradient descent using the inverse decaying learning rate schedule with initial learning rate $\eta_0 =$

0.1, $k=0.4$, where $t=1, 2, \dots$ denote the iteration index. Show the updated parameter values after each iteration. [4M]

Q5. Consider the following training data set in $\mathbb{R}^2$.

Point	x_1	x_2	class
X_1	2	2	+1
X_2	4	4	+1
X_3	2	0	-1
X_4	4	0	-1

(a) Let $w = (w_1, w_2)^T \in \mathbb{R}^2$ denote the weight vector and $b \in \mathbb{R}$ the bias term.

Formulate the hard-margin Support Vector Machine (SVM) primal optimization problem for the given dataset. [2M]

(b) show that the line $x_2 = 1$ is a separating hyper plane for the given data. [2M]

(c) show that the line $x_2 - x_1 = -1$ is a separating hyper plane for the given data. [2M]

(d) Compute the margin of each hyperplane obtained in parts (b) and (c). Determine which hyperplane provides the larger margin and justify your answer. [2M]
